Trends in Farming in the US

Written by Daniel Mastrobuono and Skyler Dooley

Overview and Motivation:
Over the past several decades, the number of farms in the United States has steadily declined, reflecting broad structural changes in agriculture, land use, and the economics of food production. While many studies examine long-term trends stretching back to the early 20th century, our project focuses on a more recent and concentrated window: the last five years. This period captures the post-COVID economic landscape, ongoing consolidation of agricultural operations, and shifting demographic patterns among farm owners.

Our goal is to understand whether the downward trend in farm numbers has continued in the most recent years and how this decline varies across states. By pairing a national-level line chart with a state-level interactive map, we aim to provide both a macro and micro perspective on modern U.S. farming. The motivation behind this project is not only to confirm whether farms are still disappearing, but also to explore the geographic patterns and potential structural forces behind these changes.

Related Work:
Our project draws inspiration from several sources that examine agricultural change and data visualization techniques. Historical USDA reports and academic analyses of farm consolidation provided context for understanding long-term trends in U.S. agriculture. We also reviewed existing visualizations of agricultural data, including interactive maps published by the USDA and examples discussed in class that demonstrated effective use of color encoding, geographic projection, and temporal comparison. These examples highlighted the value of combining national summaries with state-level detail, as well as the importance of interactivity in helping users explore regional differences.

Questions:
Our project began with a central guiding question: *Are we still losing farms every year in the United States?* This question reflects widespread assumptions about the decline of small and mid-sized farms and the consolidation of agricultural land.

As we explored the data, our questions evolved in several ways:

- *How consistent is the decline across states?*
    - We became interested in whether certain regions were more affected than others.
- *Does land in farms decline at the same rate as the number of farms?*

    - This led us to examine whether fewer farms necessarily meant less farmland, or whether consolidation was occurring.
- *Are there states that show temporary increases or stability?*
    - Identifying exceptions helped us understand the complexity of the trend.
- *What year-to-year changes are most significant?*
    - The map's change-based color encoding encouraged us to look more closely at annual fluctuations.

These evolving questions shaped both our analysis and our visualization design, pushing us to create tools that highlight differences across states and across metrics.

Data:
Our data comes directly from the USDA's *Farms and Land in Farms* annual summaries. These reports provide state-level statistics for three key metrics: number of farms, land in farms (in 1,000 acres), and average farm size. The USDA publishes these reports as PDFs, so our first step was to extract the tables and convert them into a clean CSV format suitable for analysis and visualization.

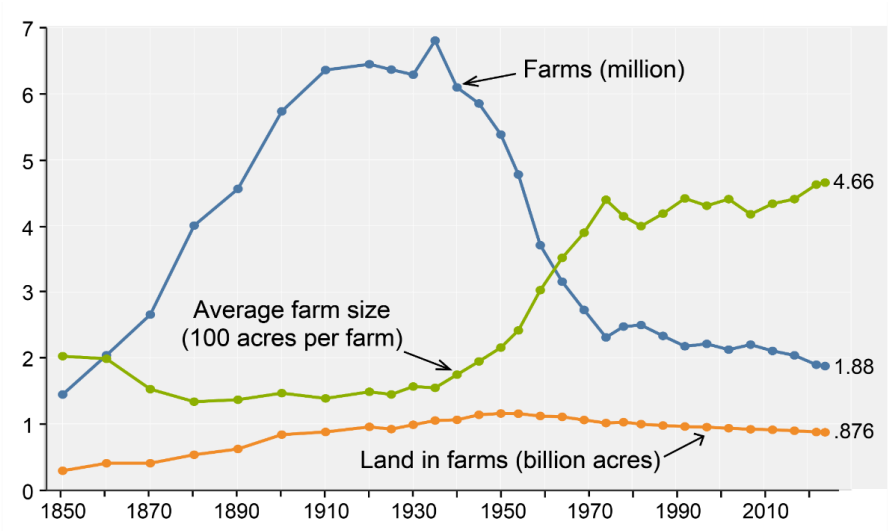
Data sources and processing steps included:
- Source PDFs: All original USDA reports used in this project are included in our code folder under /origFiles.
- Extraction and cleanup: We converted the PDF tables into CSV files, standardized column names, removed headers, and ensured consistent formatting across years. This allowed us to merge multiple years into a single master dataset.
- GeoJSON for mapping: The U.S. state boundaries used in our interactive map come from the publicly available GeoJSON file at https://eric.clst.org/tech/usgeojson/
- Final datasets: From the master CSV, we generated smaller files for the line chart (year_totals.csv) and the map (state_farm_counts.csv and land_in_farms.csv), each structured to match the needs of our D3 components.

Exploratory Data Analysis:
At first, we were focused on using a state map because we wanted to compare farm counts across the United States. We also considered other visualization options, including a treemap. As we explored these possibilities, we developed a clearer understanding of the story we wanted to communicate. Our main goal was to compare states overall and identify where farming plays a central role. This helped us realize that

a state-by-state view would best highlight those differences. Ultimately, this led us to choose the state comparison map as our final visualization.

Design Evolution:

As mentioned earlier, we considered using a treemap but found it difficult to read. It also did not align well with the story we were trying to tell. In a publication by the Department of Agriculture, we came across a chart that we found particularly interesting:

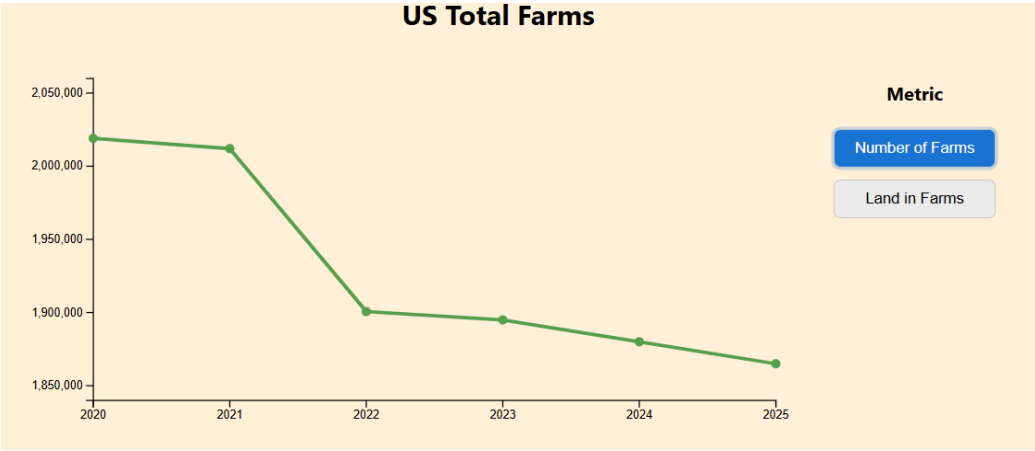


This inspired the first chart in our visualization, a high-level view of the entire US farm count. However, we thought that the y axis and confusing labels with the arrows was bad practice, so we split it into several different charts. From what we learned in the course, we aimed to make our visualization easy to understand, starting with the color scheme. We used red, green, and yellow because these colors are commonly recognized in the U.S. as representing bad, good, and something in between. Based on that convention, green indicates an increase in the number of farms from the previous year, yellow represents no change, and red signifies a decrease. We included a legend to clarify this, although the stoplight color scheme should feel intuitive to most viewers. We also centered the content on the page, following common HCI design principles to improve balance and readability. In addition, the webpage uses a parchment-style background to align with a “farm” theme. While this choice may not directly connect to a specific concept from class, we wanted the design to feel cohesive and visually reflective of the topic.

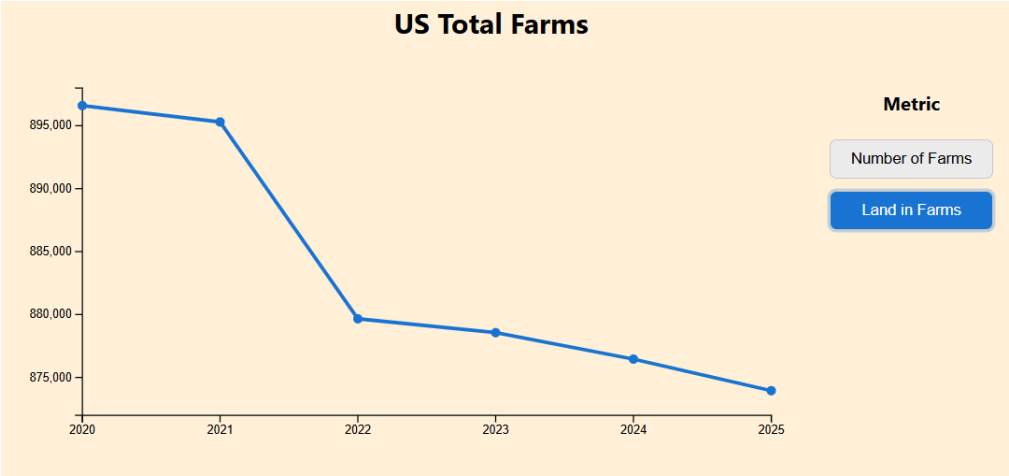
We largely stayed true to our original proposal. However, we ended up taking a more state-centric approach than we initially planned. Our goal was to determine whether the number of farms in the U.S. is still decreasing and, if possible, explore the reasons why.

Based on our data alone, we were able to identify trends but not clearly explain the causes behind them. With additional context from external research and publications, however, readers could begin to understand the many factors contributing to this decline.

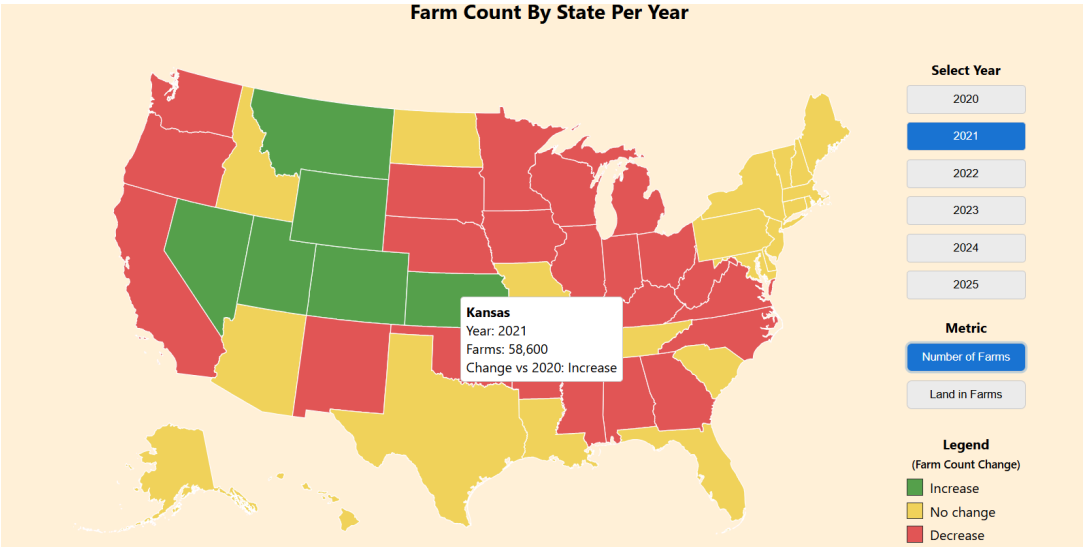
Implementation:



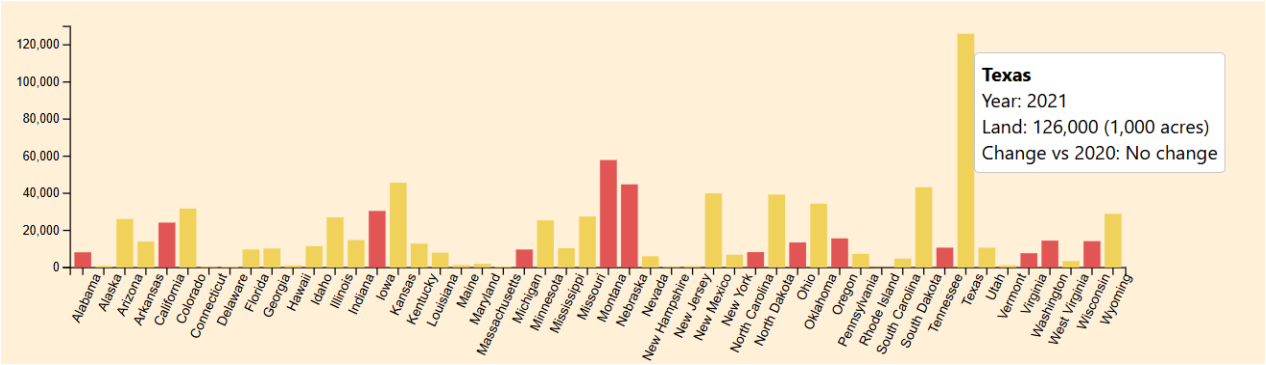
For our first graph, we aimed to keep it simple. It displays the total number of farms in the United States since 2020. This serves as the “hook” of our article, as the downward trend is immediately noticeable and clearly highlights the decline.



In terms of interactivity, one can select the metric to be displayed by the graph. The blue graph is the total farmland in 1,000 acre measurements. Even with the change in metric, the graph stays consistent with a steep downward trend between 2021 and 2022, with a more gradual trend thereafter. The purpose of this graph is to show how great of a loss in both the number of farms and land in farms throughout the US.



When the user scrolls down, they can see a US map that displays the total number of farms in each state when a state is hovered over. It also shows whether the number of farms decreased or increased from the previous year. We also provided functionality to switch between the years displayed on the graph along with a toggle for the metric to display the total farmland in 1000 acres if desired.



Finally, we included a specific bar chart for the two toggle options. This displays the same data as the map chart, but just in a different format. We wanted to add this to more easily compare the 50 states, and also in case a user was tracking one state at a time. This bar chart responds to the toggles from the previous map and the user can easily switch between the two views to get a better understanding of what was happening in each state in comparison to other states.

Evaluation:

At a surface level, the graphs clearly showed the basic trends, such as the overall change in total farms and farmland, along with more detailed state-level data over the past five years. This helped answer our original question about whether there has been a decrease both nationally and within specific states. However, when considering the "why" behind the trend, the data revealed some interesting patterns. One that stood out was the sharp decline in farms between 2021 and 2022. We speculated that this drop may have been influenced by the effects of COVID-19. It appears that the pandemic had a significant impact on farmland and farm operations. Despite that, there has not been a noticeable rebound in farm numbers since then. Instead, the U.S. has continued at this lower level without a clear recovery.

Overall, the three visualizations accomplish what we originally set out to do, but there is certainly room for improvement. For example, the tooltip on the map could be enhanced to display the exact number of farms gained or lost compared to the previous year. We could also incorporate average farm size using the original PDFs from which we sourced our data. Additionally, a scroll bar could improve the final bar graph, as the x-axis labels do not align perfectly and the layout is not as polished as it could be. Another valuable improvement would be adding sort and filter functionality to the last bar chart. Ideally, users could click on a state in either the map or the bar chart and see it highlighted in both visualizations for easier comparison. We did attempt to implement a sort and filter feature, but it proved more complex than expected. Given the time constraints and our current level of experience, we were not able to complete that enhancement.